Controller User's Manual

Ver.1.3

AIOI･SYSTEMS CO., LTD.

Revision History

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Related Manuals

- Application Development Guide
- Command Reference
- TW2208 Ethernet Junction Box Hardware Manual
- Flowlight User's Manual

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1. INTRODUCTION

1.1. Outline

This manual explains the method of connecting the controller to the PC, how to confirm the operation and communicate with the PC. It is important to read the manual before connecting the controller to the PC.

1.2. Acronyms Used in the Manual

Some of the acronyms and expressions used in the manual are shown below.

Definition

Acronyms	Explanation
JB	Junction Box
BCR I/F	Barcode Reader Interface
RS-232C I/F	RS-232C Interface

2. SETUP

This section gives an explanation on how to set up the RS-232C controller, PCI bus controller, and Ethernet controller and how they are connected to the PC.

2.1. RS-232C Controller Setup

Setup flow:

See [**2.1.1. Dipswitch Setup**] → Connect to the PC with the RS-232C straight cable → [**3. Operation Confirmation**].

2.1.1. Dipswitch Setup

Check to see that all the dipswitches located on the side of the unit are OFF (factory default setup).
Refer to the chart below if you want to change the communication format of RS-232C at default or start the controller with the special mode,



SW		OFF((Up) Factory Default Setup Mode	ON (Down)	Explanation
1	Maintenance mode setup		• Send the **A** command when power is supplied. • No retry to connected devices	When the switch is **ON**, check if the controller and light module are connected properly. Damaged light modules can be easily found. Used for maintenance and normally the switch should be **OFF**.
2	Not used			
3	Device that the controller controls the Start / End operation within a block	Signal light	Batch display	When the switch is **OFF** and an operation instruction is given to light modules within a block, a signal light with the corresponding block address will light up. When the operation is completed, the light will turn OFF. When the switch is **ON**, the batch display can be controlled in the same manner as the signal light. When the operation is completed, it will be turned OFF. However, the display instruction is sent directly from the PC.
4	Not used			
5	Not used			
6	RS-232C transmission format -- parity or no parity	Parity	No parity	Specifies parity or no parity communication format. When **OFF** (no parity), Switch 7 (SW7) is ignored.
7	RS-232C transmission format – select parity	EVEN (even number)	ODD（odd number）	When Switch 6 (SW6) is **ON**, parity setting becomes valid.
8	RS-232C transmission speed	9600bps	4800 bps	

※ Turn the switch OFF when the unit is not in use.

※ When changing the dipswitch setup, always turn the power OFF. The setup will be valid when the power is turned back on.

※ When the dipswitch is pressed down, it is ON. When the switch is up, it is OFF. This switch has to be either ON or OFF.

※ At the factory default setup, all switches are OFF (up).

2.2. PCI Bus Controller Setup

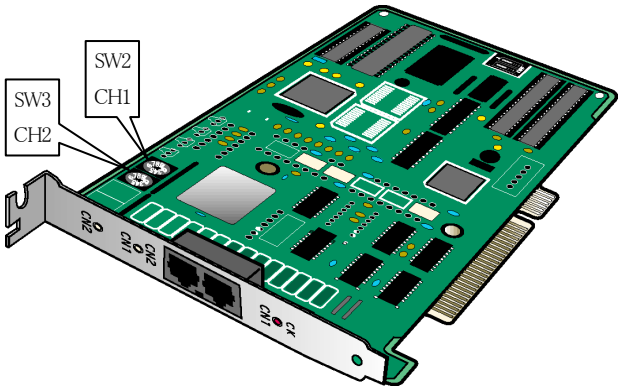
The setup flow

[**2.2.1 Digi-switch Setup**] → [**2.2.2. IoLibX Installation** (install the driver)] → [**2.2.3 Installation of the PCI Bus Controller to the PC**] → [**3. Operation Check**]

2.2.1. Digi-switch Setup

Specify the number of channels that will be used using SW2 (CH1) and SW3 (CH2) setup.
Setup the upper limit of addresses of connected light modules and other devices for each channel using the digi-switch.
The dipswitch setup must be CN1<CN2. If 9 is set up, that channel will be unused.

SW2(CHI), SW3(CH2) Setup Value				
0	:	1	~	999
1	:		~	1999
2	:		~	2999
3	:		~	3999
4	:		~	4999
5	:		~	5999
6	:		~	6999
7	:		~	7999
8	:		~	7999
9	:	Not used		



※The factory default setup is **SW2(CH1)=7,SW3(CH2) = 9. (Can only use CH1)**
※For a device with address from 9101 ~ 9199 (BCR I/F and RS-232C I/F), addresses are split according to the number of channels used.

Example 1: Factory Default Setup [SW2(CH1)=7, SW3(CH2)=9]

	Device with 1~7999	Devices with 9101 ~ 9199 addresses
CH1	1~7999	9101 ~ 9199
CH2		

Example 2: SW2 (CH1) is set up at 0, SW(CH2) is set up at 7

	Device with 1~7999	Devices with 9101 ~ 9199 addresses
CH1	1~999	9101 ~ 9150
CH2	1000~7999	9151 ~ 9199

※Normally, use the factory default setup (use CH1 only).

2.2.2. IoLibX Installation

IoLibX is AIOI's Windows-based device driver, and it must be installed when using the PCI bus controller.
This driver is stored on the CDROM. Install it before connecting the PCI bus controller to the PC.
To do the actual installation, a user with the "Administrator" authorization must perform it※.
To install, refer to the **IoLibX** Installation Manual.

2.2.3. Installation of the PCI Bus Controller to the PC

Open the PC cover and insert the PCI bus controller into an empty PCI expansion slot. If screws are used, make sure that they are screwed in tight.

Note

- When handling the controller, always be careful with respect to static electricity and shock.
- Never connect or remove the controller when the power is "**ON**".
- Never touch the gold-plated terminals. This could cause a connection failure.

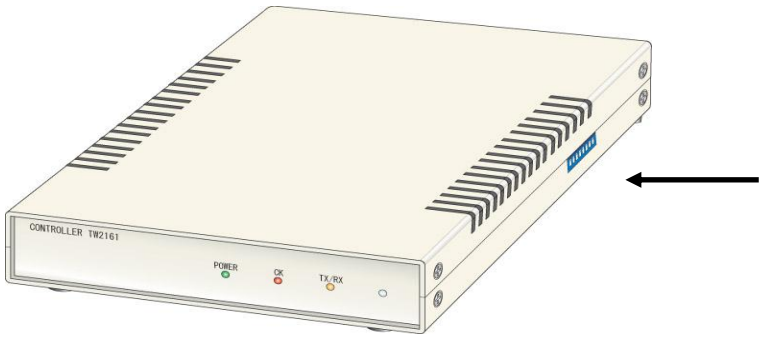
2.3. Ethernet Controller Setup

The setup flow

[**2.3.1. Dipswitch Setup**] → [**2.3.2. Factory Default Network Setu**p] → [**3. Operation Check**]

2.3.1. Dipswitch Setup

Check to see that all switches on the side of the Ethernet controller are turned OFF (factory default setup).
If special mode is to be used to start the controller, refer to the chart below for the setup.
When using the TW2208, open the cover of the Ethernet Junction Box and use the dipswitch on the Ethernet interface board.
Refer to the "TW2208 Ethernet Junction Box Hardware Manual" for details on the dipswitch position.



SW		OFF (Up) Factory Default Setup Mode	ON (Down)	Explanation
1	Mode setup for Maintenance		• Send the "**A**" command when power is supplied. • No retries to the connected devices.	When the switch is ON, checks if the controller and light modules are connected correctly and aids in finding light module malfunction. Used mainly for maintenance; therefore for normal operation, turn this switch OFF. **※ This function is applicable for the firmware version higher than 4.1.**
2	Not used			
3	Not used			
4	Not used			
5	Not used			
6	Not used			
7	Not used			
8	Not used			

※ Depending on the firmware, some functions can be applicable while other functions are not applicable.
※ For switches "Not used", turn the switches (SW) to OFF. 。
※ When changing the dipswitch setup, always turn the power of the unit OFF. The setup will be valid when the power is turned back ON.
※ When the dipswitch is pressed DOWN, it is ON. When it is UP, it is OFF. Always set to either ON or OFF.
※ At the factory default setup, all switches are OFF (UP)

2.3.2. Factory Default Network Setup

For the Ethernet controller operation, it is necessary to set up the IP Address, Subnet Mask, and Default Gateway Address according to the network environment. The chart below shows the factory default setup.

	Default Setup
IP Address	192.168.1.254
Subnet Mask	255.255.0.0
Default Gateway Address	0.0.0.0
TCP Port Number (Port Address)	5003（Fixed） **※1**

※To return to the factory default setup, refer to section「**2.3.4. Return to the Factory Default Setup**」.

※1 For controllers that can perform multi-session, a port address is fixed at 5003 and 4 variable

2.3.3. Set Up IP Address・Subnet Mask・Default Gateway

(1) Connect to the controller by using Telnet and log in.

MS-DOS prompt

```
>telnet 192.168.1.254 (Enter)
```

When the connection is successful, the login password screen is displayed.

```
login:
```

Login (log in name): **samurai**
Password: **kimono**

When the log in name and password are entered:

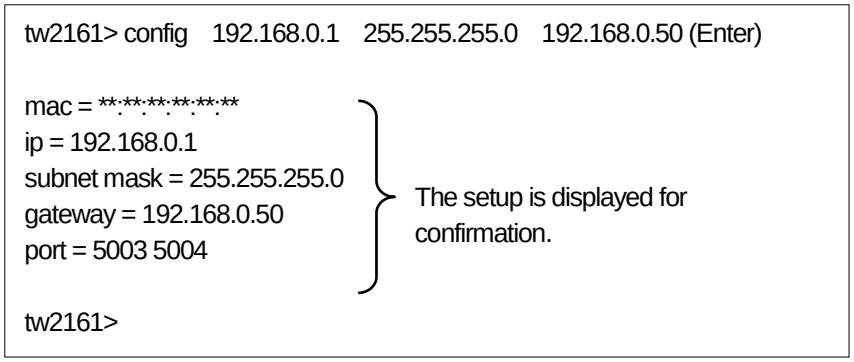
```
login: samurai (Enter)
Password: kimono (Enter)  ← Will appear with ******
tw2161>
```

(2) With the "**config**" command, enter the network information you want to set up.

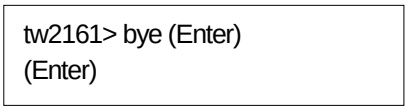
Command Format:
config <IP Address> <Subnet Mask> <Default Gateway>

Example:
IP Address = 192.168.0.1
SubnetMask = 255.255.255.0
DefaultGatewaqy = 192.168.0.50

The above setup is shown in the diagram below.



(3) With the "bye" command, logout from the Telnet.



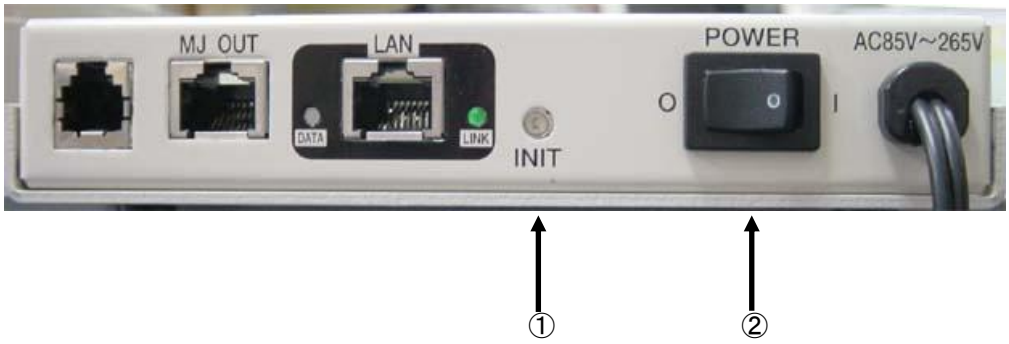
After the "bye" command, press the "Enter" key twice to log out.

(4) Turn OFF the controller's power.

When the power is turned back ON, the network setup information will take effect.
After completing the setup with Telnet, always turn the power OFF.

2.3.4. Return to the Factory Default Setup

To return to the factory default setup, turn the power (②) ON while pressing the "INIT" switch(①) located on the rear panel with a sharp point, such as a pen point. This operation is used when the controller's current setup (such as IP address) is unknown and it is not possible to reset.



In the case of the TW2208, open the cover and turn the power ON while pressing the "**INIT**" switch located on the Ethernet interface board.

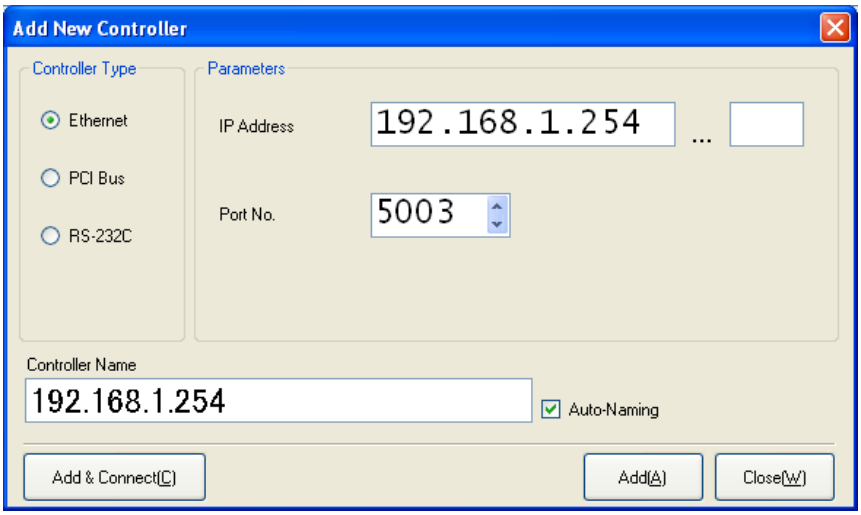
For more information on the position of the "**INIT**" switch, refer to [**TW2208 Ethernet Junction Box Hardware Manual**].

3. OPERATION CHECK

This section explains the method of performing the operation check of the controller by using, "**Flowlight**", AIOI Systems' checking tool.

3.1. To Start Flowlight

Using the CD for software development, install "**Flowlight**". Insert the CD, select "**Flowlight**" folder and select ""**Setup.exe**" to install the program.
Run "**Flowlight**" from the desktop "**Start**" menu.
The following screen will be displayed. Set up the appropriate parameters for the controller you are using.



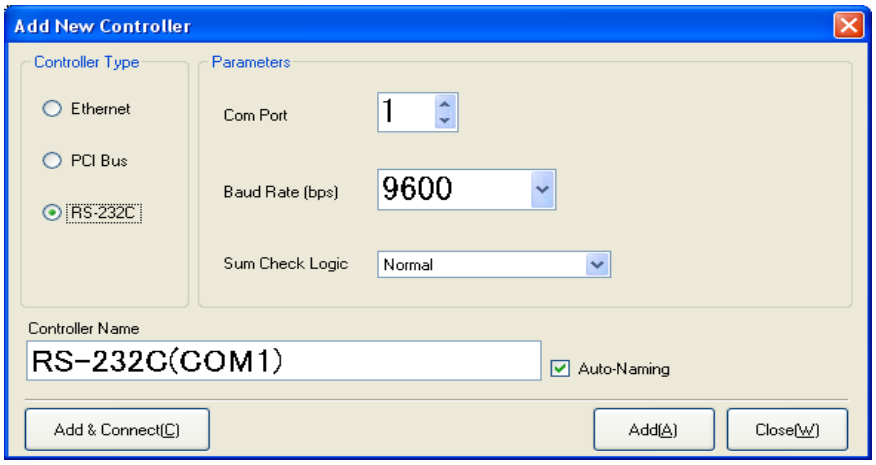
3.2. Configuration for Each Controller

After selecting the controller you are using from "**Flowlight**", a screen showing configurations for different types of controllers will be displayed. Do the necessary setups from the setup screen.

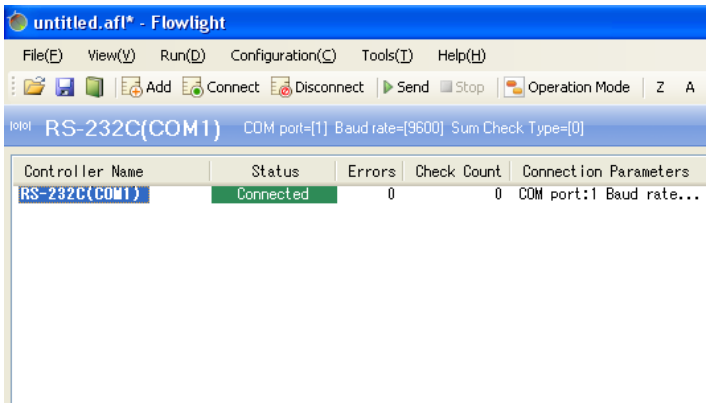
For information on the configuration of each type of controller, refer to the following sections.

3.2.1. RS-232C Controller Configuration

① With "**COM Port** ", specify the RS-232C port number of the PC connected to the controller. (The default is **COM1**).
② With "**Baud Rate (bps)**", specify the communication speed (the default is 9600bps).
③ With "**Sum Check Logic**", specify the checksum calculation method. (The default is "**Normal**").
④ Next, press the "**Add & Connect(C)**" button.



⑤ The screen switches to the main window to display the connection status.



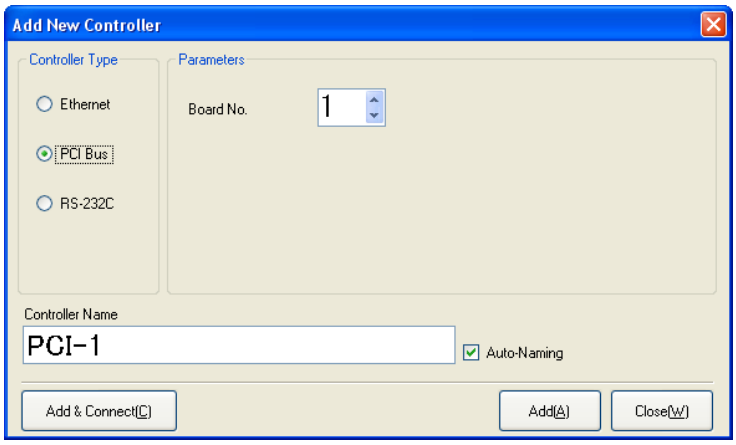
After finishing the configuration setup by following the above steps, proceed to Section [**3.3. Sending the "Z" Command**].
If "**Error**" appears after pressing the "**Add & Connect(C)**" button and you cannot open the "**COM**" port, recheck the connection between the controller and the PC.

※ For the RS-232C controller's communication setup method, refer to [**2.1.1. Dipswitch Setup**].

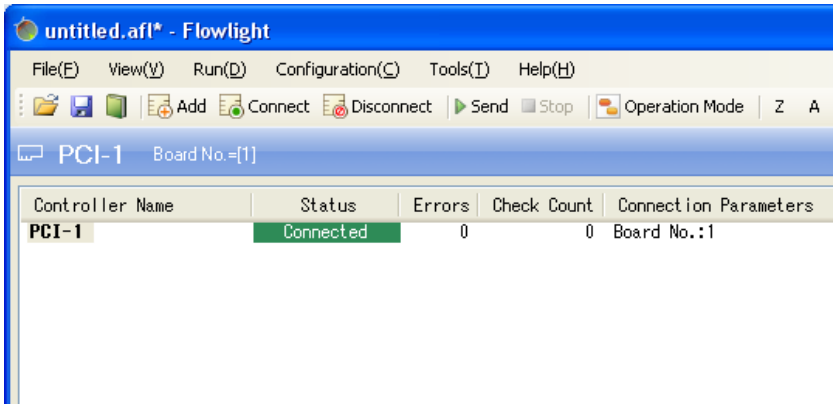
3.2.2. PCI Bus Controller Configuration

① Specify the "**Board No**" (*) used by the controller. (The default is 1).
② If you are not sure which board number to choose, temporarily close the window shown below and check the available board number(s) from the "**Tools**" menu, select "**PCI Bus Controller List**."
③ Then, press the "**Add & Connect(C)**" button.

(*)The "**Board No**" starts from 1 assigned by **IoLibX**.



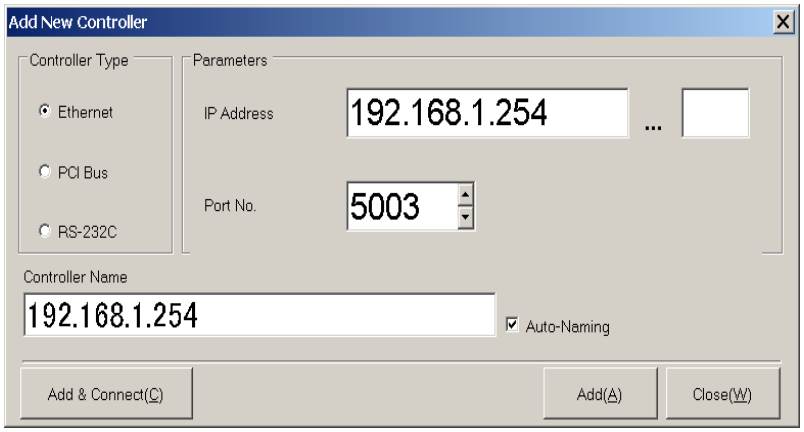
④ The screen switches to the main window to display the connection status.



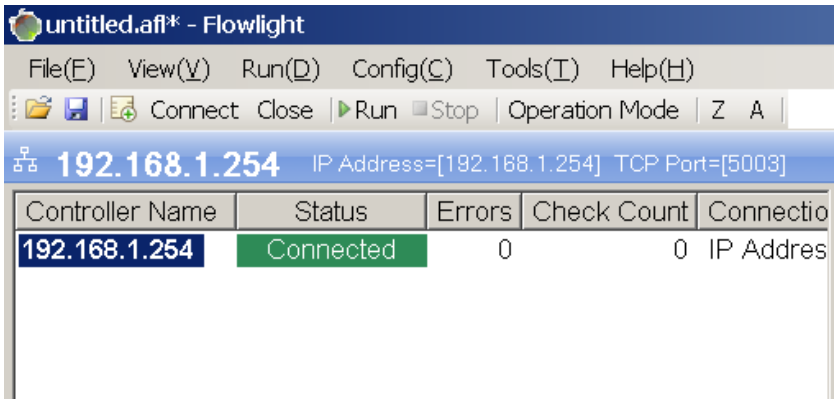
This completes the configuration setup. Proceed to section [**3.3. Sending the "Z" Command**].
If "**Disconnected**" appears after pressing the "**Add & Connect(C)**" button (③), double check to see if "**IoLibX**" (driver) has completed the installation or the board has been firmly inserted into the PC slot.

3.2.3. Ethernet Controller Configuration

① Input the IP address of the Ethernet controller in the box next to "**IP Address**"
② Enter **5003** in the box next to "**Port No.**"
③ Then, press the "**Add & Connect(C)**" button.



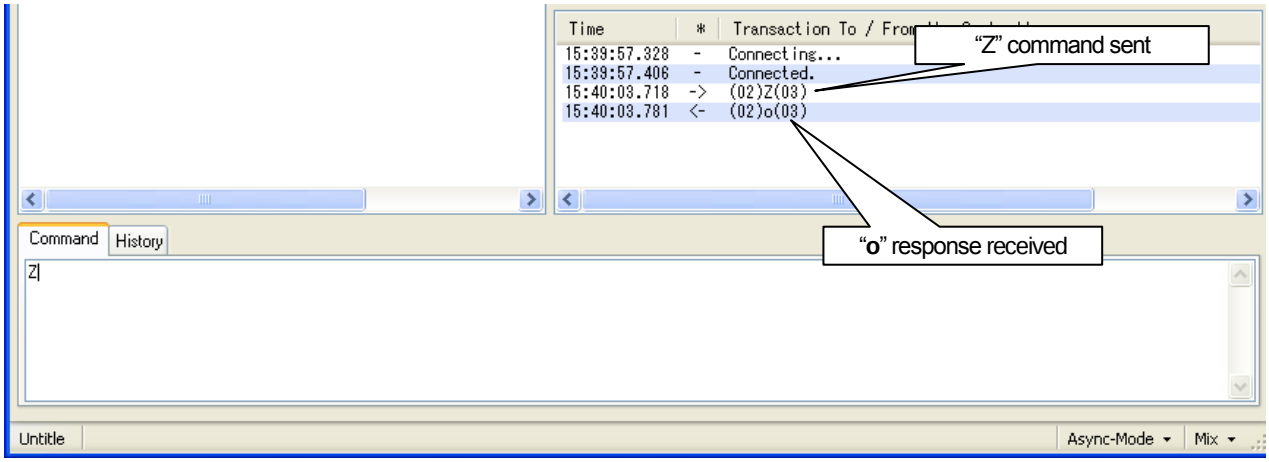
④The screen switches to the main window to display the connection status.



This concludes the configuration setup. Proceed to section [**3.3. Sending the "Z" Command**]. At this point, if "**Disconnected**" is displayed and cannot open the controller, recheck the connection between the controller and the PC.

3.3. Sending the "Z" Command

① In the command input area, input "**Z**" and press "**Enter**".
② If the response "**o**" is displayed each time "**Enter**" is pressed, the connection between the PC and the controller is operating normally without any problems.



If there is no response "o" (a message will show on the screen, for example, "The operation has timed out."), refer to section "3.2 Configuration for Each Controller" and recheck the connection between the PC and controller.

4. COMMUNICATION FORMAT AND METHOD

This section explains the format and method of communication between the PC and RS-232C controller, PCI bus controller, and Ethernet controller.

4.1. RS-232C Controller

4.1.1. Communication Format（RS-232C Controller）

Connection Standard	RS-232C
Communication Method	Full duplex system
Transmission Speed	2 options: either 9600 bps (default) or 4800bps
Character Format	Data length: 8 bit (fixed) Stop bit: 1 bit (fixed) Parity bit: 3 options: None (default), Odd, or Even

When changing the controller's transmission speed, or parity, refer to section [**2.1.1. Dipswitch Setup**]. The cable used to connect with the PC is the RS-232C straight cable.

4.1.2. Communication Frame (RS-232C Controller)

There are 4 types of communications formats for the RS-232C controller: 2 types when the communication is from the PC to the controller and 2 types from the controller to the PC.

4.1.2.1. Command Frame (PC → Controller)

Command	SM	CR

Name	Size	Details
Command	Variable length	Refer to Command Reference manual [4. Command Details]
SM (Checksum)	1 fixed byte	Use the "※Method of Checksum Calculation "shown below to calculate the "**Command**" portion, then add the value (1 byte).
CR (Frame terminating character)	1 fixed byte	Add the "**0Dh"** (1 byte) at the end of the command frame.

※Method of Checksum Calculation

① Each character of the "Command" portion is divided into two 4-bit BCD (binary coded decimal) and the BCDs are added up.

② The lower digit of the added value of ① in 4 Bits is converted to ASCII, and the 1 byte data of the converted 4-bit data is the checksum.

Example: If the "Command" portion is L00010112345,

L 0 0 0 1 0 1 1 2 3 4 5
= (4Ch) (30h) (30h) (30h) (31h) (30h) (31h) (31h) (32h) (33h) (34h) (35h)
= (4+12) (3+0) (3+0) (3+0) (3+1) (3+0) (3+1) (3+1) (3+2) (3+3) (3+4) (3+5)

= 6**6**

↓

Thus, the checksum is "**6**" (36h)

Also, the range of the checksum value is "0" (30h) ~ "9" (39h). With IOLIBX Version 3.0 and up, checksum is automatically calculated by the functions (library). The checksum is appended to the command frame and sent to the controller.

4.1.2.2. Command Response Frame (PC ← Controller)

Command Response	CR

After sending a command to the controller with the "Command Frame", the controller will always return a response (Command Response Frame) with respect to that command.

Name	Size	Details
Command Response	Variable length	"**o**" (6Fh) 1 byte ・・・ The command was executed by the controller normally. "**n**" (6Eh) 1 byte ・・・ An error in the command sent from the PC to the controller. Other ・・・ There are other commands that are returned to the PC besides "**o**", "**n**" For more information, refer to "Command Reference Manual" , 2.2 Commands Sent and Response.
CR (frame terminating character)	1 byte, fixed	Indicates the ending of the command response frame with 1-byte "**0Dh**".

4.1.2.3. Interrupt Frame (PC ← Controller)

Interrupt Data	SM	CR

The data from BCRI/F, RS-232CI/F, 10-key BCRI/F are sent from the controller to the PC using the "Interrupt Frame".
In the case of BCRI/F, for example, this frame is sent from the controller to the PC as an interrupting data after a barcode is scanned.

Name	Size	Details
Interrupt Data	Variable length	The header of the interrupt data is always 1 byte **"t"(74h)**. The header of the command response frame "4.1.2.2 Command Response Frame" will never be **"t" (74h)**; therefore, the PC can tell the difference between these 2 frames. For details, refer to the Command Reference manual, "TW Series/XW Series Command Reference, 4.13. Report from the Controller".
SM Checksum)	1 byte, fixed	The value (1 byte) obtained from calculating the checksum of the "Interrupt Data" portion by using the method described in Section "4.1.2.1, Method of Checksum Calculation" is added.
CR (frame terminating character)	1 byte, fixed	Indicates the ending of the command response frame with 1-byte "**0Dh**".

4.1.2.4. Interrupt Response Frame (PC → Controller)

Interrupt Response	SM	CR

After receiving the data from the controller with the "Interrupt" frame, send a response (Interrupt Response Frame) with respect to the data from the PC to the controller as the highest priority.

Name	Size	Details
Interrupt Response	1 byte, fixed	"**O**" (4Fh) 1 byte ··· Added when the "Interrupt Frame" from the controller is received correctly. "N" (4Eh) 1byte ··· Added when the "Interrupt Frame" from the controller is **not** received normally (e.g., received the "interrupt Frame" but the SM value did not match.) In addition, if after adding the "**N**" (4Eh), the "Interrupt Response Frame" is sent from the PC to the controller, the controller will resend the same "Interrupt frame" to the PC.
SM (Checksum)	1 byte, fixed	The value (1 byte), obtained from calculating the checksum of the "Interrupt Response" portion by using the method described in Section "4.1.2.1, Method of Checksum Calculation", is added.
CR (frame terminating character	1 byte, fixed	Add "**0Dh**" (1 byte) as the last addition to the "Interrupt Frame".

4.2. PCI Bus Controller

4.2.1. Communication Format (PCI Bus Controller)

Standard Connection	PCI bus connection
Communication Method	I/O port access (normal port monitoring)

"**IoLibX**". AIOI System Co's device driver in Windows, is used for the communication with the PCI bus controller. Refer to section [**2.2.2 "IoLibX" Installation**]. Also, for more information on how to use the "**IoLibX**" communication functions with the PCI controller, refer to "**IOLIBX.DLL Reference Manual**".

4.2.2. Communication Frame (PCI Bus Controller)

There are 4 types of PCI bus controller communication frames: 2 types from the PC to the controller and 2 types from the controller to the PC.

4.2.2.1. Command Frame (PC → Controller)

STX	Command	ETX

Name	Size	Details
STX (start of text)	1 fixed byte	Add "**02h**", 1 byte, at the beginning of the command frame
Command	Variable length	Refer to "Command Reference Manual, [4. Commands]."
ETX (end of text)	1 fixed byte	"**03h**" 1 byte is added as the end of the command frame.

4.2.2.2. Command Response Frame (PC ← Controller)

STX	Command Response	ETX

After a command is sent to the controller with the Command Response Frame, there is always a response (Command Response Frame) from the controller for that command.

Name	Size	Details
STX (start of text)	1 fixed byte	"**02h**"'s 1 byte is added as the start of the command frame.
Command Response	Variable length	"**o**" (6Fh) 1 byte ・・・ The command was executed correctly by the controller. "**n**" (6Eh) 1 byte ・・・ There is an error in the command sent from the PC to the controller. Other ・・・ There are other commands besides "**o**" and "**n**" responses that return to the PC. For more information, refer to "Command Reference Manual, " [2.2 Sending a Command and Getting a Response].
ETX (end of text)	1 fixed byte	"**03h**""s 1 byte is added as the end of the command frame.

4.2.2.3. Interrupt Frame (PC ← Controller)

STX	Interrupt Data	ETX

Data from the BCR I/F, RS-232C I/F, ten-key RS-232C I/F are sent from the controller to the PC through the "Interrupt Frame". In the case of the BCR I/F, for example, when a barcode is scanned, this frame is sent immediately as an "interrupt data" from the controller to the PC.

Name	Size	Details
STX (start of text)	1 fixed byte	"**02h**"s 1 byte is added as the beginning of the command frame.
Interrupt Data	Variable length	The head of the "Interrupt Data" is always 1 byte, "**t**" (**74h**). The 1-byte head of the command response frame of [4.2.2. Command Response Frame] can never be "**t**" (**74h**); therefore, the PC knows the difference between the two frames. For details, refer to the manual, "Command Reference," [4.13 Report from the Controller].
ETX (end of text)	1 fixed byte	"**03h**"", 1 byte, is added at the end the command frame.

4.2.2.4. Interrupt Response Frame (PC → Controller)

STX	Interrupt Response	ETX

When data is received from the controller with the "Interrupt" frame, a response for the data (Interrupt Response Frame) is sent from the PC to the controller as the highest priority.

Name	Size	Details
STX (start of text)	1 fixed byte	"**02h**"s 1 byte is added as the beginning of the command frame.
Interrupt Response	1 fixed byte	"**O**" (4Fh) 1 byte ··· Added when the "Interrupt Response" frame from the controller is correctly received. "**N**" (4Eh) 1 byte ···. Added when the "Interrupt Response" frame from the controller is not correctly received. (Received the "Interrupt frame" but did not match with the SM value, etc. . . .) Also, if after adding the "**N**" (4Eh), the "Interrupt Response Frame" is sent from the PC to the controller, the controller will resend the same "Interrupt frame" to the PC.
ETX (end of text)	1 fixed byte	Add "**03h**"" 1 byte to end the command frame.

4.3. Ethernet Controller

4.3.1. Communication Method (Ethernet Controller)

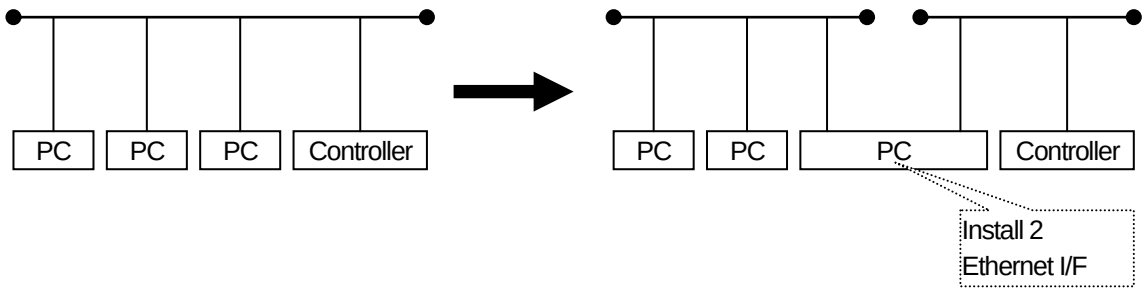
Standard connection	Ethernet 10BaseT
Communication protocol	TCP/IP
TCP/IP API	Socket Interface (Server Type)
TCP Socket Server Port No.	5003 (fixed) ※ Using the multi-session controller, can set up 5003 (fixed) and 4 variable ports.

The Ethernet controller is the Ethernet socket server. Light modules are controlled from the Client (PC) through Socket communication.
Normally, 1 socket session is used.
If several sessions are going to be established and controlled simultaneously by 1 controller, refer to section [**7. Multi-session**].
If the setup value related to the Ethernet controller network, such as the IP address, is going to be changed, refer to section [**2.3. Ethernet Controller Setup**].

Connect and Disconnect Socket from the PC ▪ Status on the Controller Operation

Status of the controller / Socket Operation of the PC	The socket connection with the PC has not been established.	The socket connection has already been established.
Socket connection	Establish session after receiving a connection request from the PC.	First, receive a connection request from the PC and establish the session. Next, disconnect the previous socket connection from the controller side.
Socket disconnection		Receive a request to disconnect from the PC, then disconnect the session

Note: The network structure that includes the Ethernet controller
A high demand is placed on the Ethernet controller to operate in real time, and for this reason, excessive communication load from devices unrelated to the operation of the Ethernet controller may cause delayed response. In some situation, for devices that receive heavy traffic, such as PC server and host computer, it is advisable to put them on a completely separate network.



4.3.2. Communication Frame (Ethernet Controller)

There are 4 types of Ethernet controller communication frames – 2 types from the PC to the controller and 2 types from the controller to the PC.

4.3.2.1. Command Frame (PC → Controller)

STX	Seq No.	Data Length	Command	ETX

Item	Size	Details
STX (start of text)	1 fixed byte	"**02h**", 1 byte, is added at the start of the command frame.
Seq No. (Frame Sequence No.)	3 fixed bytes	The range of the "**SeqNo.**" value is 000 -999, and normally it starts from 000. Each time the controller sends a command, the value increases by 1. If the value reaches 999, the next value returns to 000. **General rule: "SeqNo." increases by 1 (+1) each time a command is sent**. 000 -> 001 -> 002 ・・・・ 998 -> 999 -> 000 -> 001 ・・・ The value of the "**SeqNo.**" is always 3 digits, "0" is used to fill in the missing digits.
Data Length	4 fixed bytes	Specifies the length of the "Command" portion in bytes. The range of the value of data length is 0001 - 1015. For example: If the "Command" portion is 5 bytes Data length becomes 0005 If the "Command" portion is 120 bytes Data length becomes 0120 The value is always 4 digits. "0" is used to fill in the missing digits.
Command	Variable length	Refer to "Command Reference Manual" [4. Commands]
ETX (end of text)	1 fixed length	"**03h**"", 1 byte, is added at end the command frame.

4.3.2.2. Command Response Frame (PC ← Controller)

STX	Seq No.	Data Length	Command Response	ETX

After sending a command to the controller using the Command Frame, the controller will always return a response (Command Response Frame) with respect to the command.

Name	Size	Details
STX (start of text)	1 fixed byte	"**02h**", 1 byte, is added at the start of the command frame.
Seq No. (Frame sequence No.)	3 fixed bytes	The same value as that of the command frame "**Seq No.**" is added.
Data length	4 fixed bytes	The length of the "Command Response" portion in bytes. The length is always 4 digits. Use "0" to fill in the spaces in front when less than 4 bytes.
Command response	Variable length	"**o**" (6Fh) 1 byte ···The command was correctly executed by the controller. "**n**" (6Eh) 1 byte ···. The command sent from the PC to the controller had an error. "**n1**" (6Eh + 31h) 2 bytes ··· The "**Seq No.**" of the command sent from the PC to the controller is the same "**Seq No.**" as the previous command frame. If this happens, commands sent from the PC will no longer be processed. Normally, this type of response will not be returned if the command frame Seq. No. rule is followed. Other Commands ··· there are other commands besides "**o**", "**n**", "**n1**". Refer to "Command Reference" manual, [2.2. Sending a Command and Getting a Response].
ETX(end of text)	1 fixed byte	"**03h**"", 1 byte, is added at end the command frame.

4.3.2.3. Interrupt Frame (PC ← Controller)

STX	Seq No.	Data Length	Interrupt Data	ETX

Data from BCRI/F、RS-232CI/F、10-key RS-232CI/F and information after the **CONFIRM** button was pressed are sent by the "Interrupt" frame from the controller to the PC.
For example, in the case of BCRI/F, when a barcode is scanned, this frame is sent immediately from the controller to the PC as an "Interrupt Data".

Normally, in the case of the RS-232C and PCI bus controller, the PC itself sends the "**H**" command to get the status after pressing the **CONFIRM** button. In the case of Ethernet controller, however, this information is sent to the PC with the "Interrupt Frame".

Name	Size	Details
STX (start of text)	1 fixed byte	"**02h**"s 1 byte is added as the beginning of the command frame.
Seq No. (Frame sequence No.)	3 fixed bytes	Starts from "000" and each time the "Interrupt frame" is sent from the controller to the PC, the sequence number increases by 1. Returns to 000 after 999. Be aware that this sequence number is appended spontaneously and has no relationship with the "**Seq No.**" of the "Command Frame • Command Response Frame"
Data length	4 fixed bytes	The length of the "Interrupt data" (in bytes) Must always be 4 digits. Use "0" in front when less than 4 digits.
Interrupt Data	1 fixed byte	The first byte of the Interrupt data length is always "**t**" (**74h**). Because the first byte of the command response frame, [4.3.2.2. Command Response Frame] will never be "**t**" (**74h**), the PC knows the difference between the two frames. For details, refer to the manual, "**Command Reference**," [**4.13 Report from the Controller**].
ETX (end of text)	1 fixed byte	Add "**03h**"' 1 byte to end the command frame.

4.3.2.4. Interrupt Response Frame (PC → Controller)

STX	Seq No.	Data Length	Interrupt Response	ETX

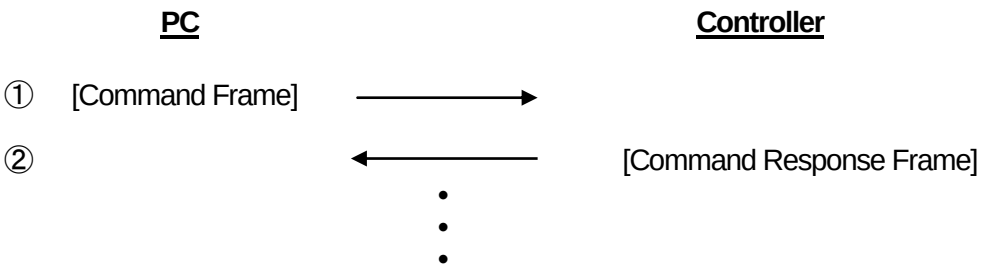
When data is received from the controller with the "Interrupt Frame", a response for the data ("Interrupt Response Frame") is sent from the PC to the controller as the highest priority.

Name	Size	Details
STX (start of text)	1 fixed byte	"**02h**"s 1 byte is added as the beginning of the command frame.
Seq No. (Frame sequence No.)	3 fixed bytes	Add the same value as the "**Seq No.**" of the interrupt frame.
Data length	4 fixed bytes	Specify the length of the "Interrupt Response" portion (1 byte). The range of the data length is 0001 − 1015. Must always be 4 digits. Use "0" in front when less than 4 digits.
Interrupt Response	1 fixed byte	"**O**" (**4Fh**), 1 byte ••• Added when the interrupt frame from the controller was correctly received by the PC. ※With Ethernet controller, "**N"** (**4Eh**, Resend) cannot be specified for the interrupt response.
ETX (end of text)	1 fixed byte	Add "**03h**"' 1 byte to end the command frame.

4.4. Communication Protocol

Explained below is the communication protocol between the PC and controller.
There may be some differences among various controllers; however, the basic protocol is the same.

I. Sending a command from the PC to the controller



① Put the command on the [Command Frame] and send it from the PC to the controller.

② The PC must wait at least 3 seconds to receive the [Command Response Frame] from the controller (5 seconds in the case of the Ethernet controller. (※1) (Receive time out)

There is no need to wait after the PC receives a response (※2). If there is another command that you want to send, repeat ① - ②.

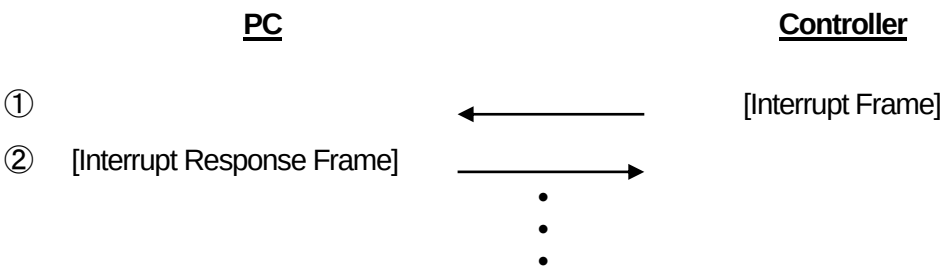
Do not send the next command to the controller before receiving the [Command Response Frame] for the command sent earlier.

Normally, the controller will return the [Command Response Frame] to the PC, but due to some external physical problems, such as disconnection between the PC and controller, a response may not be returned within the specified time. For mission-critical system, it is important to have a way to recover from the communication problem. Refer to [**4.5. Recovery from Communication Failure**] and take necessary preventive steps.

※**1**) The PC will receive a response from the RS232-C controller or PCI bus controller within 3 seconds and 5 seconds (for the Ethernet controller. However, it is necessary to wait longer when there are unusual light modules, commands used, or the number of specified light module addresses. For more information, refer to "**Command Reference**" on the command processing time.

※**2**) For example, in the case of the Ethernet controller, if a command is sent from the PC to the controller and the response took 1 second, there is no need to wait for the remaining 4 seconds.

II. Protocol of PC Response to the Controller after Receiving the "Interrupt Frame" from the Controller



① When the barcode scanned, the scanned information is sent by an "Interrupt Frame" from the controller to the PC.

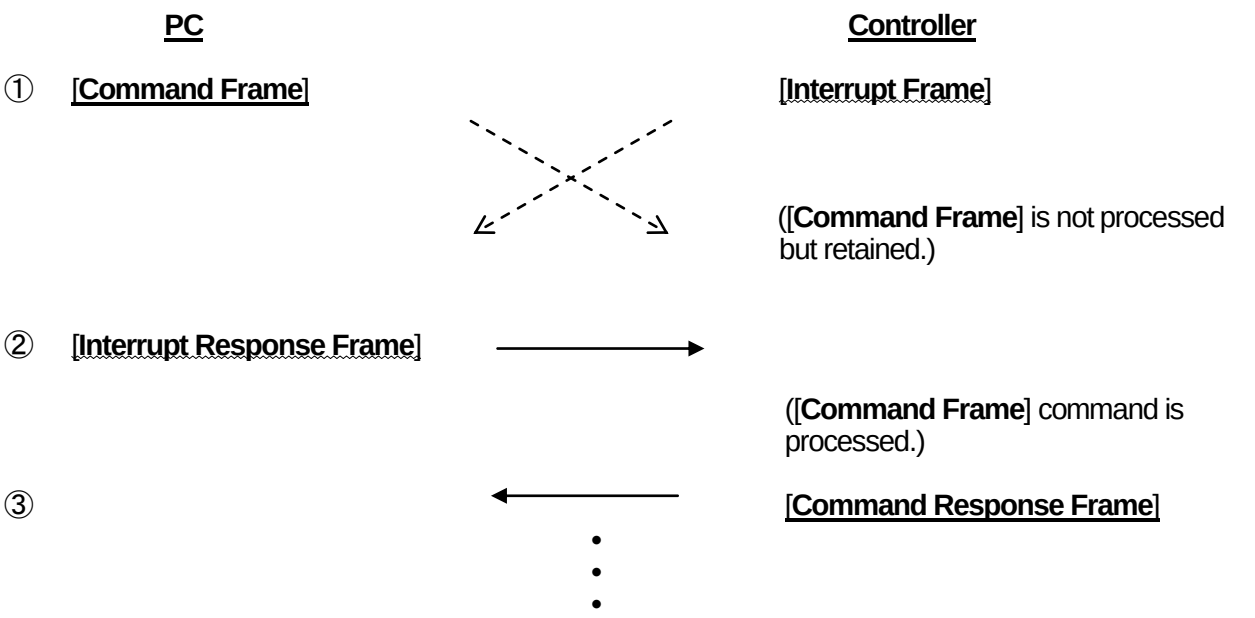
② As soon as the PC receives the "① frame", immediately (**※1**) send a response, "Interrupt Response Frame" to the controller.

After sending the "Interrupt Frame" to the PC, the Ethernet controller will receive the "Interrupt Response Frame" from the PC within 5 seconds. If the response is not received, the "Interrupt Frame" will be resent. When it is resent, PC should respond with the "Interrupt Response Frame". (The RS-232C and PCI bus controller will not resend, but will wait until the "Interrupt Response Frame" is received from the PC.)

※1) After the controller sends the "Interrupt "Frame" to the PC, it cannot process a new command from the PC until it receives the "Interrupt Response Frame" from the PC. Do not send a new command until the PC has been allowed the required time to send the "Interrupt Response Frame".

If the PC receives the "Interrupt Frame" (Section I - ②) while waiting for the "Command Response Frame" from the controller, " the response for the "Interrupt Frame" must be done with high priority. For details, refer to section III.

III. Communication Protocol When the Transmission of "Command Frame" from the PC and the "Interrupt Frame" from the Controller Overlap



① The PC was waiting for a response after sending the "Command Frame", but the frame received from the controller was "Interrupt Frame".

② The PC does not wait for the response for the "Command Frame", but sends the "Interrupt Response Frame" to the controller.
At this point, the controller retains the command from the PC (①) and is at the "Interrupt Response Frame" receiving stage. It then starts processing the command.

④ After processing the command, the controller returns the "Command Response Frame" for the "Command Frame" (①)

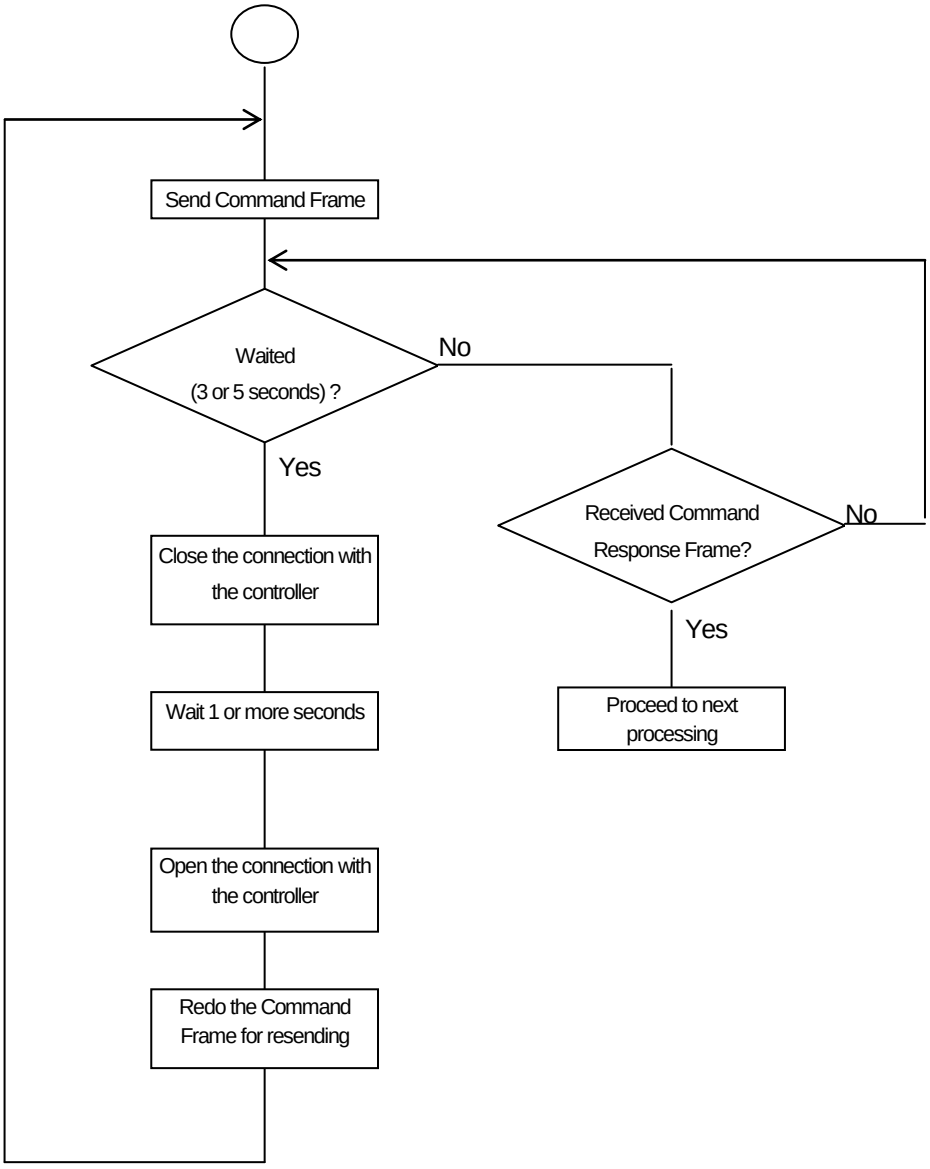
Regarding the above, in order for the controller to quickly send the "Command Response Frame" to the PC, a series of commands from "Interrupt Frame" to "Interrupt Response Frame" must be processed rapidly.

4.5. Recovery from Communication Failure (Resend)

The PC sent the command, "Command Frame", but there was no response, "Command Response Frame", from the controller after waiting 3 or more seconds※ due to the communication problem between the PC and controller. When this happens, close the connection with the controller. Then, reopen and resend the "Command Frame".

※For Ethernet controller, wait 5 or more seconds.

[Flow Chart of Resend]



Precaution When Redoing the Command Frame for Resend

① For the Ethernet controller, use the last Command Frame's "**Seq No.**"**+1** as the "**Seq No.**" of the resend frame. (If over 999, make it 000).

② Beside the overwritable block operation instruction, such as the "**P5**" command, be careful to ensure there is no duplicate resend of the operation instruction commands, such as the "**L**," "**LP1**", "**P4**", etc.,

Example:
Creating the command frame that takes into account of "duplicate sending":

- Replace with "**P5**" command. At this point, each operation mode is specified "No Change".

What to Watch for When Resending to Reprocess:

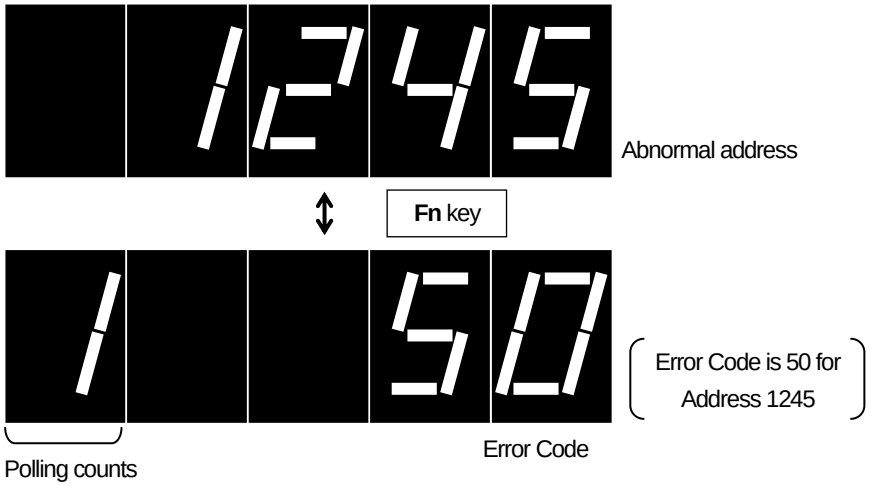
① When a light module malfunction happens, the standard waiting time (3 seconds for the RS232-C and PCI bus controllers or 5 seconds for the Ethernet controller) needs to be longer depending on which commands are used and how many light module addresses have been specified in the commands. For details, refer to "**Command Reference**" manual, [**Section 2.3 Waiting Time for a Response**].

② Allow more than 1 second for the controller to close and open.

③ If an error occurs when the controller is closing and opening, the cause is usually due to some physical damage, such as a bad cable; therefore, check the connection between the PC and controller.

5. NUMBER 0 LIGHT MODULE

Number "0" (zero) light module is usually a 5-digit light module with "0" address setting. It can be connected to the junction box.
When the controller receives the operation instruction or "Open" of the barcode reader interface through the PC, it monitors the addresses. If it determines that the address is abnormal, the address and error code are outputted to the number 0 light module.



5.1. Error Code Table of Number 0 Light Module

Code	Details
1	Stop for an instant
2	BCC error after retry
50	No response (disconnected or damaged)
51	No response (framing or overrun error)
7	• More than 2 operation instruction commands are sent to the same light module (redundant transmission) • Operation instruction commands cannot be received.

Connection Method:
Connect to the junction box. It can also be connected to a channel not yet set up to use.
Make sure that the address of Number 0 light module is "0" (zero).

Operation Method:
Any malfunction will be displayed on the Number "0" light module. There will be a beeping noise and the LED will flash.
At first, the address of the abnormal light module will be displayed. When the "**Fn**" key is pressed, the polling count and error code will be displayed. When the "**Fn**" key is pressed once more, the address will be redisplayed. When the **CONFIRM** button is pressed, the display is cleared. The Number "0" light module cannot retain malfunction information of multiple light modules. It can only display information on the most recent abnormal light module. If the Number "0" light module continues to make the beeping noise even after pressing the **CONFIRM** button to clear the data, there may be multiple abnormal light modules. When that happens, monitor the Number "0" light module for a while.

6. LOG FILES (For Ethernet Controller only)

The Ethernet controller records and retains 3 types of operation information as log files.
Use these log files because they contain useful debugging information on a system that uses this controller.

6.1. Number "0" Light Module Log File (Number 0 Log)

The controller stores the data outputted to the Number "0" light module as a log file ("Number 0 log").
The Number "0" light module can only display the most recent error, but the "Number 0 log" is added from the time power is supplied. Also, "Number 0 log" can be downloaded through FTP. (※)

※ Do not download the "Number 0 log" during the picking operation (while waiting for a response after sending a command such as "L" and "P1").

(1) Connect with FTP and get the log file.

Connect with the controller using FTP and enter the following:
User name: **samurai** Password: **kimono**
and download "**modlog_a.txt**" or "**modlog_b.txt**"

When power is supplied, "**modlog_a.txt**" is added, but the maximum size of 1 log file is limited to 4,096 bytes. If a file is larger than 4,096 bytes, "**modlog_a.txt**" is copied to "**modlog_b.txt**" and a new file "**modlog_a.txt**" is created.

Example of Getting "Number 0 log" with the MS-DOS Prompt

```
C:¥>ftp 192.168.1.254    ← Using FTP connect to the controller
Connected to 192.168.1.254.
220 FTP Server ready
User (192.168.1.254:(none)): samurai
331 Password required
Password: kimono      ← Password will not be displayed
230 Logged in
ftp> dir              ← Internal file is displayed for checking
200 PORT Command successful
150 Opening data connection
modlog_a.txt      4096
modlog_b.txt      332
226 Transfer complete
ftp: 44 bytes received in 0.05Seconds 0.88Kbytes/sec.

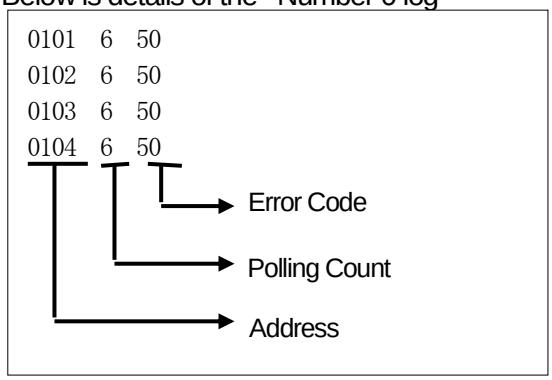
ftp> get modlog_b.txt c:¥modlog_b.txt  ← Put the log file in PC's C drive
200 PORT Command successful
150 Opening data connection
226 Transfer complete
ftp: 18 bytes received in 0.16Seconds 0.11Kbytes/sec.
ftp> bye    ← Terminate FTP
221 Goodbye
```

The format of the "**Number 0 Log**" file is the following:

＜Address＞ ＜Polling Count＞ ＜Error Code＞＜Line Feed＞

Refer to Section [**5.0 Number 0 Light Module**] for an explanation of various values.

Below is details of the "Number 0 log"



6.2. Error Retry Counter (Error Retry Counter Log)

The controller continuously counts the number of retries due to the communication error with devices. It counts each time the module encounters a problem and can be shown using the "**twlog**" command via Telnet. Also, the file can be downloaded using **FTP**. (**See Note**)

Note: Do not attempt to display or download "**Error Retry Counter** " during the picking operation (e.g., while waiting for a response after sending the operation instruction command, such as the "**L**" or "**P1**" command.)

6.2.1. Error Retry Counter Display

(1) Connect with **Telnet** and log in.

From MS-DOS prompt, etc. , Enter the following:

```
>telnet 192.168.1.254  (Enter)
```

When successfully connected, the login screen appears.

```
login:
```

Login (log in name): **samurai**
Password: **kimono**

```
login: samurai (Enter)
Password: kimono (Enter)  ← Displays "*" for the password.
```

Input the command, "**twlog list**" to display the number of errors counted.

Command Format:
twlog list

```
tw2161> twlog  list (Enter)

0101  1
0102  8
0212  12
```

Error counter value
Module address

If there is no "**Error Retry**" in any of the module, nothing will be displayed.

```
tw2161> twlog   list (Enter)
tw2161>
```

6.2.2. Get "Error Retry Counter" using FTP

(1) Log in by connecting to Telnet
Refer to section 6.2.1 (1).

(2) Input the "**twlog save**" command

Command Format:

twlog save

```
tw2161> twlog   save (Enter)
tw2161>
```

By executing this command, the controller converts the "**Error Retry Counter**" into a log file internally. Always execute this command in order to obtain the most recent retry information.

(3) Get the log file by connecting to FTP.

Using FTP to the controller, then enter:

User name: **samurai** Password: **kimono**

and get a file called "**twlog.txt**".

Example of obtaining a log file by MS-DOS prompt

```
C:¥>ftp 192.168.1.254    ← Use FTP to connect with the controller
Connected to 192.168.1.254.
220 FTP Server ready
User (192.168.1.254:(none)): samurai
331 Password required
Password: kimono      ← Password will not appear.
230 Logged in
ftp> dir              ← Displays the internal file for confirmation.
200 PORT Command successful
150 Opening data connection
cmdlog_a.txt      333
twlog.txt         18
226 Transfer complete
ftp: 44 bytes received in 0.05Seconds 0.88Kbytes/sec.

ftp> get twlog.txt c:¥twlog.txt  ← The log file is saved in the PC's C drive.

200 PORT Command successful
150 Opening data connection
226 Transfer complete
ftp: 18 bytes received in 0.16Seconds 0.11Kbytes/sec.
ftp> bye    ←  Quit FTP
221 Goodbye
```

The content of the log file (**twlog.tx**) has the same format as the format displayed by "**twlog list**" (Telnet).

6.2.3. How to Clear the "Error Retry Counter"

The "Error Retry Counter" can be cleared when the controller's power supply is turned OFF or with the command, "**twlog clear**" using Telnet.

The command shown below clears the "Error Retry Counter".

(1) Connect with Telnet and log in.

Refer to Section 6.2.1 (1).

(2) Enter the command, "**twlog clear**"

Command Format:

twlog clear

```
tw2161> twlog  clear (Enter)
tw2161>
```

The "Error Retry Counter" is cleared by executing this command.

6.3. Command Log File

The controller stores the content of the commands sent to / from the Client as a log file.
This file can be downloaded using FTP. **(※)**

※ Do not download a log file during the picking operation, e.g., while waiting for a response after sending the "L" and "P" commands or other operation instruction commands.

(1) Connect to FTP and get the log files.

To connect to the controller using FTP, input:
User name: **samurai** Password: **kimono**
After connecting, get the log files: "**cmdlog_a.txt**" and / or "**cmdlog_b.txt**".

When power is supplied, the log will be added to "**cmdlog_a.txt**"; however, the maximum size of 1 log file is 4,096 bytes. If the log file is larger than this, "**cmdlog_a.txt**" is copied to "**cmdlog_b.txt**", and a new file, "**cmdlog_a.txt**" is created.

Shown below is how to get the "Command Log" with MS-DOS prompt.

```
C:¥>ftp 192.168.1.254    ← Connect to the controller using FTP
Connected to 192.168.1.254.
220 FTP Server ready

User (192.168.1.254:(none)): samurai

331 Password required

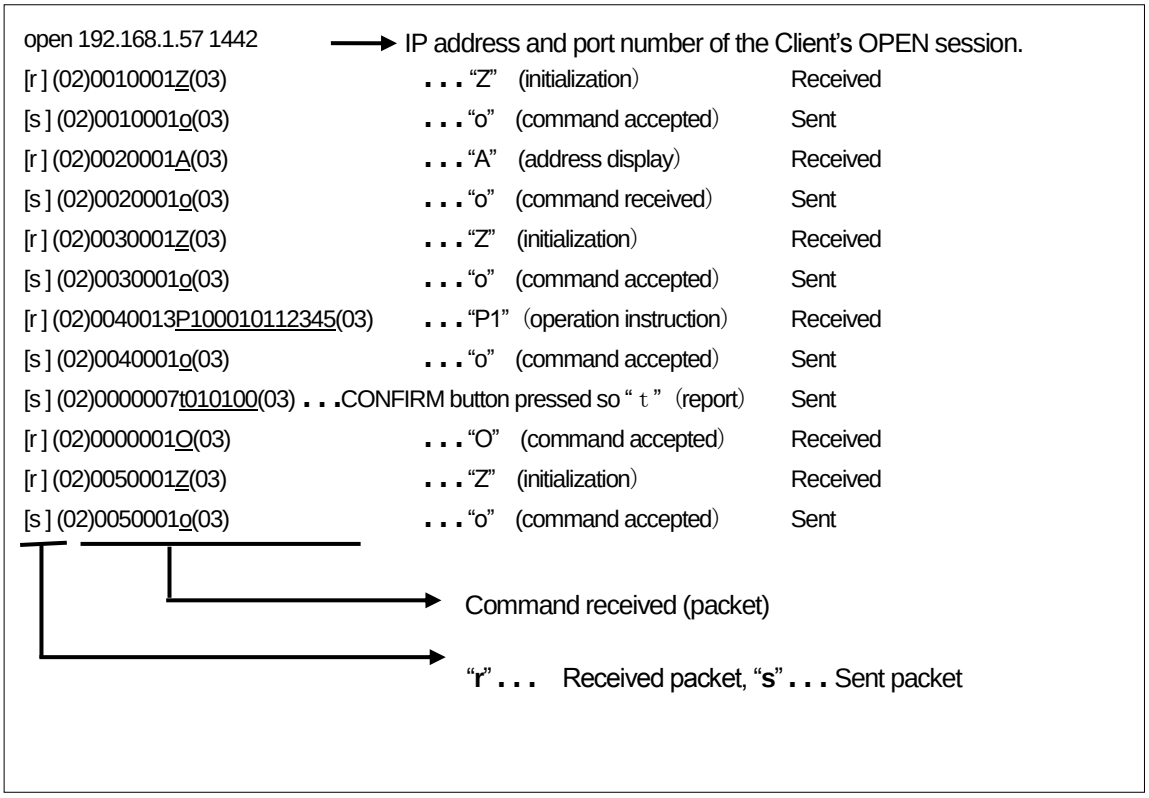
Password: kimono        ← This password will not be displayed.

230 Logged in
ftp> dir                ← For confirmation, the internal file will be displayed.
200 PORT Command successful
150 Opening data connection
cmdlog_a.txt        4096
cmdlog_b.txt        332
226 Transfer complete
ftp: 44 bytes received in 0.05Seconds 0.88Kbytes/sec.

ftp> get cmdlog_b.txt c:¥cmdlog_b.txt  ← The log file is saved in the PC's C drive.

200 PORT Command successful
150 Opening data connection
226 Transfer complete
ftp: 18 bytes received in 0.16Seconds 0.11Kbytes/sec.
ftp> bye    ← Quit FTP
221 Goodbye
```

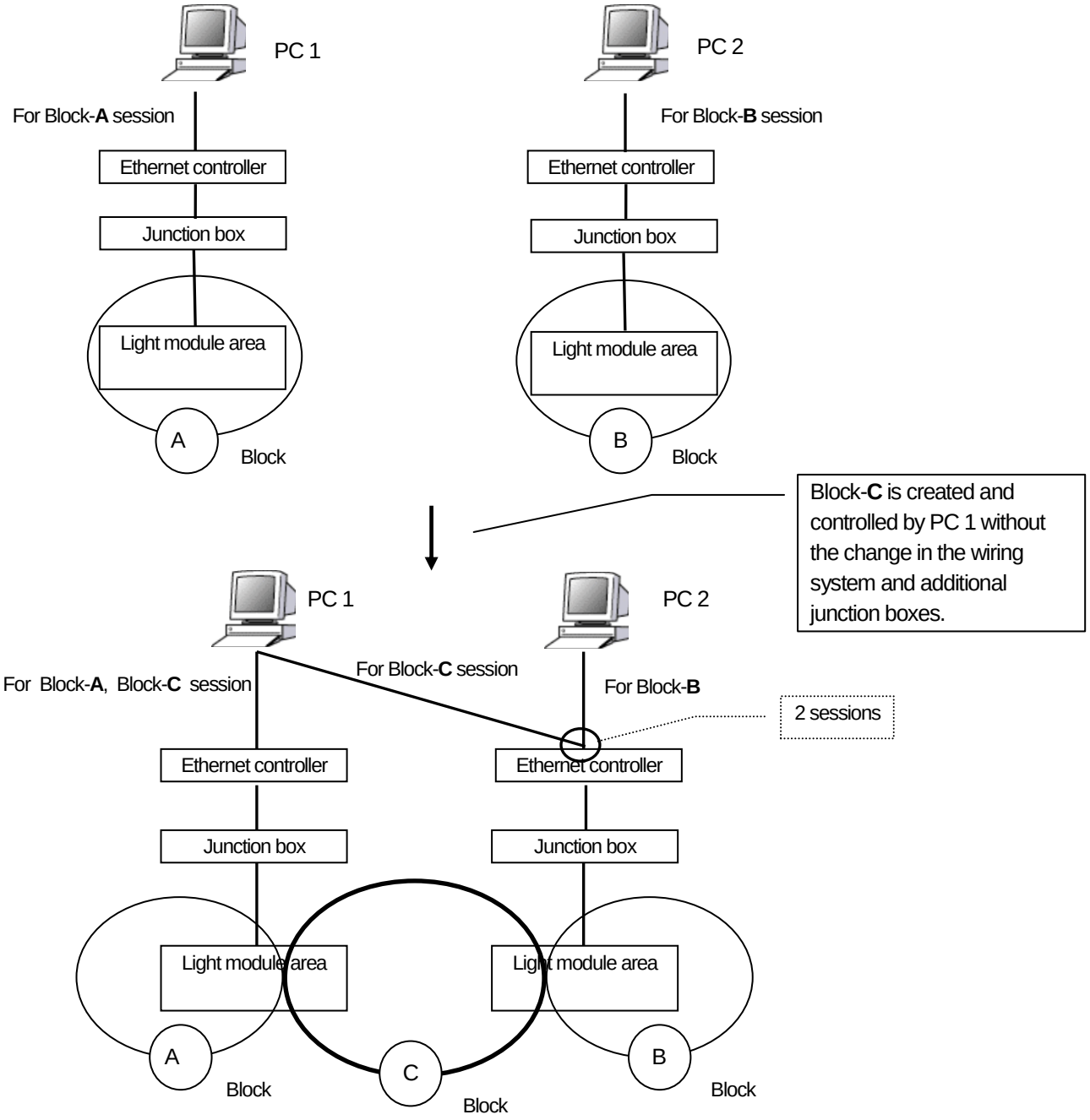
A sample "Command Log" is shown below.



7. MULTI-SESSION

The controller that can perform multi-session has a maximum of 5 ports, and it can receive sessions (TCP/IP socket) at the same time.
For the controller's reception ports, since 1 port cannot receive several sessions, the controller is equipped with several different port numbers. It is, therefore, necessary to select a port for the control PC. Within the maximum 5 ports, 1 port has a fixed port number, **5003**. The port numbers of the remaining 4 receiving ports can be changed by the user.

Structure:



※Regarding what to watch for when using various control commands on multi-sessions, refer to Section [**7.3. What to Watch for When Using Commands**].

7.1. Controlling Sessions

Looking at 1 port = 1 session, the method of controlling a session is the same as the controller that can only control 1 session.
If the controller connects to a port (stretched session) before the previous session is completed (not yet terminated), the controller will force it to terminate. (Refer to **4.3.1. Communication Method, Connect and Disconnect Sockets from the PC – Status of the Controller Operation**). If that happens, there will be no impact on the sessions of other ports.

7.2. Setup and Change of the Receiving Port Number

At default, the receiving port number is 5003. If the user wants to change and set up port numbers, connect via Telnet and do the setup using the special command. Below is the setup method.

(1) Connect to Telnet to log in.

Refer to Section 6.2.1 (1).

(2) Using the "**set if**" command, enumerate the receiving port numbers of the controller.

Command Format:
set if – PortNumber1 PortNumber2 PortNumber3 Portnumber4

When all 4 ports are to be set up: (Example, 4051, 4052, 4053, 4054)

```
tw2161> set if –port 4051 4052 4053 4054 (Enter)

tw2161> bye
```

When only 2 ports are to be set up: (Example, 4051,4052)

```
tw2161> set if –port 4051 4052 (Enter)

tw2161> bye
```

This completes the setup. However, in order to reflect this to the controller, it is necessary to restart the controller by turning the power back ON.

Warning **The specification of ports for multi-session is different depending on the version of the controllers.**

(1) Controller version < 6.0

5003 is fixed.

The user can change the setup up to 4 ports.

(2) Controller version ≧ 6.0

The user can set up a maximum of 5 ports.

※Refer to section 7.4 on how to check the version of the controller.

To confirm the content of the setup, use the command: "**show if**".

The figure below is a display example for controller version < 6.0.

```
tw2161> show if (Enter)

Physical Address(MAC) . . . . . : **:**:**:**:**:**
IP Address. . . . . : 192.168.1.254
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 0.0.0.0
Application Port(fixed) . . . . . : 5003
Application Port(1) . . . . . : 4051
Application Port(2) . . . . . : 4052
Application Port(3) . . . . . : 0
Application Port(4) . . . . . : 0
```

The receiving ports are 3 ports: 5003, 4051, and 4052. Ports (3) and (4) are "0", indicating they are not yet setup. These 2 additional ports can be set up in the future.

The figure below is an example for controller version ≧6.0.

```
tw2161> show if (Enter)

Physical Address(MAC) . . . . . : **:**:**:**:**:**
IP Address. . . . . : 192.168.1.254
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 0.0.0.0
Application Port(1) . . . . . : 4051
Application Port(2) . . . . . : 4052
Application Port(3) . . . . . : 4053
Application Port(4) . . . . . : 4054
Application Port(5) . . . . . : 4055
```

The receiving ports are 5 ports: 4051, 4052, 4053, 4054, and 4055.

Warning **Duplicate multi-session port numbers are not allowed.**

The user can set up any port number, but if the same port number is used to set up multiple ports, there will not be normal communication with the controller. To set up multiple ports, use different port numbers.

7.3. What to Watch for When Using Commands

Be careful with the command that sends data to all modules and the command specified with wildcard. Below are some examples:

Examples:

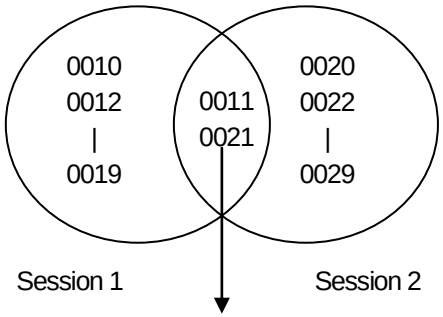
① The controller sends the initialization command, "**Z**" to all modules controlled by the controller. If another "**Z"** command is sent from a separate session even in the midst of the operation and the light on the session port is illuminated, the light module in operation will turn OFF unexpectedly.

The module will start initializing when power is turned ON without sending the "**Z**" command and can receive various commands. However, if it is necessary to send the "**Z**" command, timing must be taken into consideration. (The "**Z**" command can be sent when there are no other sessions or when the picking operation has been completed with other sessions.)

② The "**A**" command is for maintenance. The same caution applies here as in ①.

③ The address conflict from using the wildcard between different sessions must be avoided.

Be careful when using a wildcard. As an example, suppose for Session 1, the addresses are 0010 to 0019 and for Session 2, 0020 to 0029, and a wildcard is "00?1" is used, the wildcard will affect both 0011 of Session 1 and 0021 of Session 2.



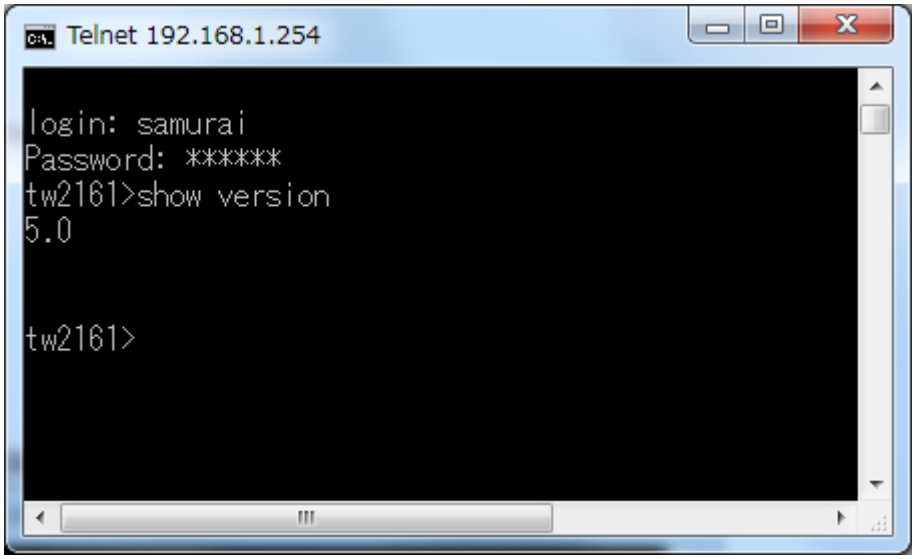
An overlapped address portion from the wildcard

7.4. How to Check the Controller Version

(1) Connect with Telnet and log in.
Refer to 6.2.1 (1).

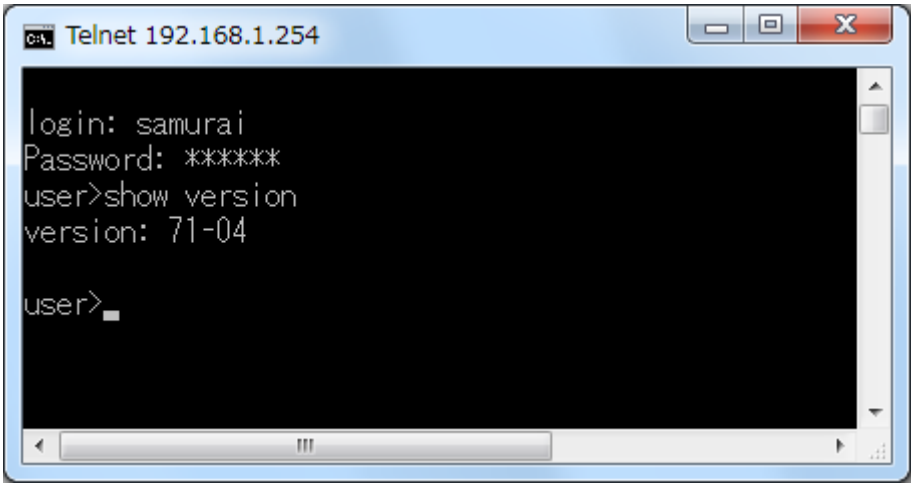
(2) Use the "**show version**" command to check the controller version.

The screen below shows the version (controller version < 6.0).



The version shown above is 5.0.

The screen below shows the version (controller version ≧ 6.0).



The above screen shows version 7.1.

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[**Inquiries**] Please send any inquiries regarding the contents of the specifications and manual of light module series, or application program development, by FAX or E-mail to AIOI·SYSTEMS CO., LTD.: info1@hello-aioi.com

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